Deep Read LLM Tokenisation

Qi Zao Lim

May 2026

1 Architecture A: Transformer - Token Dynamics

1.1 Forget BIT, It is All about TOKEN

- **Architecture Studied**: Transformers, Mamba, and Diffusion (LLaDA)
- **Token-level Object**
  Semantic information content (entropy, directed information, rate-distortion)
- **Analytical Tool**: Information Theory
- **Layer Wise vs Global**
  Global. The analysis characterizes the probabilistic model of the architectures holistically across pre-training, post-training, and inference phases.
- **Key Finding about Token Dynamics**
  Tokens (not bits) are the fundamental unit of semantic information. LLMs act as discrete-time channels, and the semantic information flow (directed information density) naturally progresses and accumulates over the token sequence.
  (Simpler Explanation) AI processes language in meaningful chunks rather than raw data. It delivers these chunks one by one, and with every new word it adds, the total "meaning" of the message grows more specific and detailed.
- **Cross-architecture Implications**
  Broadly applicable. By defining structure-agnostic information-theoretic measures, the framework is designed to evaluate Transformers, Mamba, and Diffusion models under a unified semantic umbrella.
- **Limitations**
  The framework focuses heavily on high-level mathematical probability models and global information boundaries rather than tracking the step-by-step geometric or layer-wise evolution of tokens inside the hidden states.

1.2 Bridging the Dimensional Chasm

- **Architecture Studied**: Transformers
- **Token-level Object**
  Position in the manifold and intrinsic dimension
- **Analytical Tool**: Geometry and empirical probing
- **Layer Wise vs Global**: Layer-wise
- **Key Finding about Token Dynamics**
  While transformers employ high dimensional embedding (10^3), effective model tend to compress tokens into approximately 10-dimensional submanifolds, closely resembling human semantic spaces.
  This work not only advances LLM interpretability by reframing Transformers layers as projectors

that mediate between high-dimensional computation and low-dimensional semantics, but also provides practical tools for model diagnostics that do not rely on task-specific evaluations

- **Cross-architecture Implications**
  Currently architecture-specific. As noted in your gap analysis, it remains entirely unknown if selective state spaces (Mamba) or iterative denoising processes (Diffusion) exhibit this same dimensional expansion and collapse.

- **Limitations**
  The geometric analysis relies on the explicit residual stream and layer-by-layer structure of Transformers; architectures without discrete layer-wise attention mechanisms were not evaluated.

1.3 Attention Sinks and Compression Valleys in LLMs are Two Sides of the Same Coin

- **Architecture Studied**: Transformers

- **Token-level Object**
  Token norm (massive activations) and representational rank

- **Analytical Tool**: Empirical probing, linear algebra, and information theory

- **Layer Wise vs Global**: Layer-wise

- **Key Finding about Token Dynamics**
  Proves a theoretical connection showing that massive token activations in the residual stream cause both attention sinks and compression valleys. Proposes a 3-phase "Mix-Compress-Refine" theory of how tokens flow through layers.
  (Simpler Explanation)
  Attention Sink is when AI directs a disproportionate amount of its "focus" (attention weights) to the very first few tokens in a sequence. Even if these initial tokens lack semantic meaning. **Compression Valley** is where intermediate representations show unexpectedly low entropy(ordered and predictable) despite the model's high-dimensional space.
  Proposed 3 Phase Pattern: In the first few layers, the AI is "messy." It looks at every word in your prompt and tries to see how they all relate to each other. It's gathering all possible context.
  In the middle layers, the AI gets serious. It realizes it doesn't need to keep every tiny detail in its "short-term memory." It squishes (compresses) the information into a dense, high-energy summary. This is the "Compression Valley."
  Selective Refinement (The Plating): In the final layers, the AI takes that dense summary and polishes it. it stops looking at the big picture and focuses purely on which specific word should come next.
  This framework provides a more mechanistic account of how LLM allocate compute across depth.

- **Cross-architecture Implications**
  Highly architecture-specific. Attention sinks are a byproduct of global $O(n^2)$ self-attention and specific LayerNorm dynamics. It is unknown how representational rank evolves with depth in sequential models like SSMs or routed models like MoEs.

- **Limitations**
  Phenomenon is intrinsically tied to pre-norm Transformer architectures. Models that process tokens non-autoregressively or without explicit attention matrices are outside the scope of this theory.

1.4 Entropy-Lens: Unocovering Decision Strategies in LLMs

- **Architecture Studied**: Transformers

- **Token-level Object**
  Shannon entropy of intermediate token distributions

- **Analytical Tool**: Information Theory
- **Layer Wise vs Global**: Layer-wise
- **Key Finding about Token Dynamics**
  Residual Streams are simply the sum of the output of all previous layers and the original embedding. It is akin to a communication channel, since it doesn't do any processing itself and all layers communicate through it. It allows model to have multiple layers while remembering the important things that happen in each layer (like keeping a version history)
  Introduces "Entropy-Lens" to map token dynamics in the residual stream to a 1D "entropy profile." Reveals that models use depth-invariant expansion and pruning strategies that are distinct to specific model families.
  (Simpler Explanation) "Entropy Lens" is used to analyze AI's high dimensional and categorical dynamics of token space. It shows that LLMs use expansion(consider different possibilities) and pruning(removing unlikely words) between layers.However, each model family have different processes on how they mix expansion and pruning.
- **Cross-architecture Implications**
  Highly generalizable. The primary contribution of this tool is its ability to highlight family-specific computation patterns, making it a direct comparative tool between Transformers, SSMs, and others.
- **Limitations**
  While it effectively identifies what the entropy profile looks like for different families, it is primarily an observational/diagnostic tool and may not fully explain the mechanistic why behind those entropy shifts.

1.5 MTM: A Multi-Scale Token Mixing Transformer for Irregular Multivariate Time Series Classification

- **Architecture Studied**: Transformers
- **Token-level Object**
  Clustering behavior and relative positions in representation space.
- **Analytical Tool**: Dynamical systems (Mean-field interacting particle model)
- **Layer Wise vs Global**: Layer-wise
- **Key Finding about Token Dynamics**
  When viewed as interacting particles, tokens exhibit a specific sequential evolution: a fast initial representation collapse, followed by the formation of distinct clusters, which then sequentially merge as the tokens pass through deeper layers. This paper proves the mathematical bounds of such transitions.
- **Cross-architecture Implications**
  Architecture-specific. Self-attention behaves naturally like a particle interaction system. As highlighted in your survey plan, it is an open question whether sequential processing (Mamba) causes clustering, and the concept is practically non-applicable to Diffusion LLMs where tokens are intentionally corrupted and denoised rather than clustered.
- **Limitations**
  The mathematical proofs heavily rely on the all-to-all dense interactions of self-attention. The model does not account for state-mediated processing, sparse routing, or diffusion trajectories.

1.6 The Mean-Field Dynamics of Transformers

- **Architecture Studied**: Transformers
- **Token-level Object**
  Token position, clustering, and angular dispersion on the unit hypersphere ($\mathbb{S}^{d-1}$)

- **Analytical Tool**: Geometric / Manifold and Dynamical Systems (mean-field limits and collapse theory)
- **Layer Wise vs Global**: Layer-wise. It tracks the asymptotic trajectory of tokens as layer depth approaches infinity.
- **Key Finding about Token Dynamics**
  When modeling the pure self-attention mechanism mathematically, tokens are subjected to an inescapable, attractive geometric force. Because the softmax attention matrix averages values, tokens constrained to a hypersphere rapidly lose their angular dispersion. As they propagate through successive layers, this dynamic inherently drives all tokens in a sequence to collapse into a single, degenerate cluster. Essentially, **without counter-forces (like MLPs or specific residual structures), the tokens inevitably lose their distinct semantic identities and blend into an indistinguishable mass**.
  Softmax attention matrix is the "weight" each token in a sequence pays to every other token. It dynamically extracts contextual information by producing a probability distribution of token relationships.
- **Cross-architecture Implications**
  Architecture-specific. The proof explicitly targets the mathematical properties of the softmax self-attention matrix. As highlighted in your survey gaps, it remains unknown if the sequential processing in Mamba or iterative steps in Diffusion suffer from an analogous mathematical collapse.
- **Limitations**
  This is a purely theoretical analysis of the attention mechanism in isolation. It does not account for how diverse training data, varying parameter scales, or modern architectural interventions (like MoE routing) delay or mitigate this collapse in practice.

1.7 You Do Not Fully Utilize Transformer's Representation Capacity

- **Architecture Studied**: Transformers
- **Token-level Object**
  Routing decisions and representation separability
- **Analytical Tool**: Routing / Gating and Information-Theoretic (focusing on preserving token information capacity)
- **Layer Wise vs Global**: Layer-wise processing with global memory access.
- **Key Finding about Token Dynamics**
  Standard deep Transformers force tokens into an information bottleneck; to pass context forward, a token must compress all historical data into its current layer's residual stream, which directly fuels representation collapse. By introducing a mechanism where tokens dynamically route their queries to historical states across all previous layers (on a per-head basis), tokens are freed from this bottleneck. They no longer have to carry the burden of past context continuously, allowing their current layer representations to remain highly distinct, separable, and expressive even in extremely deep networks
- **Cross-architecture Implications**
  Currently specific to Transformers, though it conceptually mirrors the "selective forgetting/remembering" goal of Mamba's input-dependent gating.
- **Limitations**
  The paper focuses on resolving token collapse but does not deeply address how cross-layer routing scales computationally regarding memory bounds or extreme context lengths. **This paper suggests improvements to existing infrastructure**.

1.8 REFORM : Residual Filtering through Neural Aggregators for Layer-Wise Representation Integrity

- **Architecture Studied**: Transformers
- **Token-level Object**
  Token separability and hidden state uniformity
- **Analytical Tool**: Geometric/Manifold and Routing/Gating
- **Layer Wise vs Global**: Layer-wise (integrating states across depths)
- **Key Finding about Token Dynamics**
  Tokens naturally suffer from severe feature homogenization (over-smoothing) as they pass through successive self-attention blocks, gradually losing their unique semantic features. REFORM introduces a hierarchical hidden state integration mechanism that acts as a structural regularization. By explicitly aggregating earlier layer outputs into current layers, it forces token trajectories to remain distinct within the manifold rather than blending into a polysemantic average. This ensures that fine-grained, token-specific information survives the journey to the final output layer.
- **Cross-architecture Implications**
  While tested on Transformers to solve attention-induced smoothing, the concept of multi-layer aggregation could theoretically benefit any deep architecture prone to representational degeneration.
- **Limitations**
  The study is isolated to Transformer baselines; it is untested on SSMs or MoE architectures to see if their inherent structural differences naturally resist the over-smoothing REFORM aims to fix. **This paper suggests improvements to existing infrastructure**.

1.9 Flowing Through Layers: A Continuous Dynamical Systems Perspective on Transformers

- **Architecture Studied**: Transformers
- **Token-level Object**
  Continuous trajectory and perturbation decay (norm)
- **Analytical Tool**: Dynamical Systems (ODE formulation, contractive dynamics)
- **Layer Wise vs Global**: Layer-wise (but modeled as continuous time/depth)
- **Key Finding about Token Dynamics**
  The discrete, step-by-step layer updates in a Transformer can be rigorously mapped to the solution of a continuous Ordinary Differential Equation (ODE). Through this lens, token dynamics are mathematically proven to be contractive. This means that as tokens move "forward in time" (deeper into the network), any initial perturbations(small, external change in input data), noise, or minor variances in their embeddings rapidly decay. While this contractive pull explains why deep Transformers are incredibly robust and resilient to input noise, it also explains their inherent vulnerability to representation collapse, as fine details are mathematically smoothed away.
  Simpler explanation: As token moves through layers, they "converge" towards a single mathematical point, meaning they have lost their individual identities. By using ODEs, we can map how tokens will act downstream.
  Representation Collapse: model maps diverse, distinct inputs into nearly identical internal representations. When the "contractive pull" is too strong, the tokens don't just get close; they become identical.Model is "dead" as every token in sequence is seen as the exact same concept, making it impossible to generate a coherent or nuanced response.

- **Cross-architecture Implications**
  Highly cross-architectural. It establishes a direct, mathematical equivalence in fundamental token expressivity between the Transformer and Mamba families.

- **Limitations**
  Circuit complexity bounds prove what an architecture can compute in theory, but they offer no guarantees about learnability—meaning it does not address whether Mamba can learn these token computations as efficiently or with the same sample complexity as a Transformer during actual training.

2 Architecture B: Mamba/SSM - Token Dynamics

2.1 The Computational Limits of State-Space Models and Mamba via the Lens of Circuit Complexity

- **Architecture Studied**: Mamba/SSM

- **Token-level Object**: Computability / sequence processing capacity

- **Analytical Tool**: Computational Complexity (Circuit complexity)

- **Layer Wise vs Global**: Global. It characterizes the absolute theoretical limits of entire architecture family

- **Key Finding about Token Dynamics**
  There is a radical mechanistic difference in how these architectures process tokens—Transformers use all-to-all dense interactions, while Mamba relies on sequential, state-mediated updates. However, this paper mathematically proves that both architectures fall into the exact same computational complexity class (TC^0 with polynomial precision). This means that despite Mamba tokens being processed one by one through a constrained state space, the architecture suffers no fundamental theoretical disadvantage in expressivity; it is capable of computing the exact same class of functions over a token sequence as a full self-attention mechanism

- **Cross-architecture Implications**
  Highly cross-architectural. It establishes a direct, mathematical equivalence in fundamental token expressivity between the Transformer and Mamba families.

- **Limitations**
  Circuit complexity bounds prove what an architecture can compute in theory, but they offer no guarantees about learnability—meaning it does not address whether Mamba can learn these token computations as efficiently or with the same sample complexity as a Transformer during actual training.

2.2 A Theoretical Analysis of Mamba's Training Dynamics: Filtering Relevant Features for Generalization in State Space Models

- **Architecture Studied**: Mamba/SSM

- **Token-level Object**: Feature alignment and input-dependent gating

- **Analytical Tool**: Dynamical Systems (analyzing continuous training dynamics and sample complexity).

- **Layer Wise vs Global**: Global (focuses on how the model evolves holistically during the training process).

- **Key Finding about Token Dynamics**

- **Cross-architecture Implications**
The defining feature of Mamba is its input-dependent gating—unlike older fixed-state models, Mamba decides what to remember based on what it is reading. This paper proves mathematically that as Mamba trains, its gating dynamics explicitly align with the "class-relevant" features embedded within the tokens. The model learns to selectively filter out background noise while amplifying critical semantic signals, dynamically adjusting its state space boundaries to capture the most important information. Furthermore, the paper provides non-asymptotic bounds proving Mamba can learn these complex, token-aligned gating functions efficiently from data.

- **Limitations**
First, the theoretical analysis focuses on a simplified Mamba setting that abstracts away practical components such as depth, multiple heads, residual connections, and layer normalization. Second, the data model, while standard in theoretical studies, also simplifies real-world sequence structures. Extending the analysis to more realistic multi-layer and multi-head Mamba architectures, richer data models, and alternative designs such as gated Transformers or hybrid Mamba–Transformer frameworks remains absent from current work.

2.3 Understanding Input Selectivity in Mamba: Impact on Approximation Power, Memorization, and Associative Recall Capacity

- **Architecture Studied**: Mamba/SSM

- **Token-level Object**: Token memory retention and information decay

- **Analytical Tool**: Routing / Gating and Information-Theoretic (memory constraints)

- **Layer Wise vs Global**: Layer-wise

- **Key Finding about Token Dynamics**
Standard recurrent networks and basic SSMs suffer from exponential memory decay; old tokens simply fade away. This paper mathematically maps Mamba's core selective mechanism (the S6 layer) to Haar wavelet projections. By dynamically adjusting these wavelet projections on a per-token basis, Mamba actively counteracts that natural exponential decay. It acts as a sophisticated, multi-scale filter that can deliberately hold onto specific, highly relevant historical tokens for massive context windows, while aggressively forgetting useless intermediary tokens

- **Cross-architecture Implications**
It highlights how Mamba achieves "routing-like" temporal selectivity, which is the sequential equivalent of a Transformer's explicit ability to look back at any past token via attention.

- **Limitations**
While the Haar wavelet projection is a brilliant theoretical mapping, it is a mathematical abstraction that may not perfectly capture the chaotic, non-linear interactions of a fully trained, multi-layer Mamba model in practice.

2.4 LaTIM: Measuring Latent Token-to-Token Interactions in Mamba Models

- **Architecture Studied**: Mamba/SSM

- **Token-level Object**: Pairwise interaction strength between tokens

- **Analytical Tool**: Information-Theoretic and Dynamical Systems (decomposing sequential states).

- **Layer Wise vs Global**: Layer-wise

- **Key Finding about Token Dynamics**
Mamba processes tokens sequentially by squashing them into a compressed 1D hidden state, seemingly destroying the direct token-to-token maps seen in Transformers. However, LaTIM proves you can

mathematically "unroll" this sequential process to extract exact, fine-grained pairwise interaction scores. The key finding is that Mamba implicitly forms highly structured dependencies over time—dynamically prioritizing recent tokens and highly salient distant tokens—without ever actually computing a full $O(n^2)$ matrix.

- **Cross-architecture Implications**
  This provides the exact mathematical tools needed to extract an "attention map" from an SSM, allowing for direct, apples-to-apples comparisons between Mamba's sequential interactions and a Transformer's global attention.

- **Limitations**
  Reconstructing these pairwise interactions from a compressed state is an intensive analytical exercise; it is an observational tool for researchers rather than a lightweight process that can be used dynamically during the model's forward pass.

2.5 The Hidden Attention of Mamba Models

- **Architecture Studied**: Mamba/SSM

- **Token-level Object**: Implicit attention weights instantiated by the token stream

- **Analytical Tool**: Routing/Gating and Dynamical Systems

- **Layer Wise vs Global**: Layer-wise

- **Key Finding about Token Dynamics**
  Mamba does not merely approximate attention. It generates "hidden" attention matrices on the fly. The data-controlled linear operators in Mamba act as micro-routers.
  Simpler Explanation Mamba belongs to a family of AI called "Selective State-Space Models." Older state-space models applied the exact same math to every piece of incoming data. Mamba is "selective"—its mathematical filters (the linear operators) change dynamically based on the specific word it is looking at right now (data-controlled). This allows Mamba to get the primary benefit of the attention mechanism(contextual information) without having to calculate the giant grid. The attention is "hidden in the math of the state space model.
  The paper reveals a staggering fact: a standard Mamba model effectively utilizes three orders of magnitude more hidden attention matrices than a comparably sized Transformer has explicit attention heads. Instead of a few global heads looking at everything, Mamba processes tokens through a massive, distributed ensemble of highly localized, implicit attention routers. Thus, by replacing a few giant, slow attention mechanisms with a massive army of highly efficient, dynamic "micro-routers," Mamba can process information incredibly fast without losing the ability to understand deep context.

- **Cross-architecture Implications**
  This completely blurs the architectural lines. It suggests that data-dependent SSMs and Transformers are not as functionally different as their code implies; Mamba is simply executing a highly factored, sequential form of self-attention.

- **Limitations**
  "Hidden attention" is an implicit mathematical property. Because Mamba cannot physically jump back to explicitly retrieve a raw token from the past like a Transformer, it still struggles with exact retrieval tasks (like the needle-in-a-haystack problem) despite these hidden matrices.

3 Diffusion LLM — Token Dynamics

3.1 Large Language Diffusion Models

- **Architecture Studied**: Diffusion LLM

- **Token-level Object**: Masking probability and bidirectional context confidence
- **Analytical Tool**: Information-Theoretic (Iterative denoising and forward/reverse probabilities).
- **Layer Wise vs Global**: Global (However, instead of analyzing spatial layers, the analysis tracks tokens across temporal denoising timesteps)
- **Key Finding about Token Dynamics**
Tokens in LLaDA completely break the standard autoregressive (left-to-right) lifecycle. Instead of accumulating meaning layer by layer, tokens are processed through forward masking (intentional corruption) and reverse prediction (iterative denoising). During inference, all tokens are updated simultaneously. The model continuously refines its confidence across the entire sequence bidirectionally, updating the probability distribution of each individual token based on the gradually emerging global context. This allows it to achieve competitive reasoning performance with strong autoregressive models like LLaMA3-8B, but via a radically different mechanism. For demo, view : https://ml-gsai.github.io/LLaDA-demo/
- **Cross-architecture Implications**
It implies a different perspective on how LLM are derived.
- **Limitations**
Because the architecture is so new, the deep mechanistic dynamics of the tokens are still largely a black box. It is currently unknown whether tokens in a Diffusion LLM cluster, collapse, or traverse low-dimensional submanifolds the way Transformer tokens do during these denoising steps.

3.2 Simple and Effective Masked Diffusion Language Models

- **Architecture Studied**: Diffusion LLM
- **Token-level Object**: Token masing probability and ELBO(Evidence Lower Bound)
- **Analytical Tool**: (ELBO optimization, cross-entropy bounds)
- **Layer Wise vs Global**: Global (tracking tokens across temporal denoising steps)
- **Key Finding about Token Dynamics**
Traditional autoregressive models learn token distributions by explicitly calculating exact next-token probabilities. Diffusion models, however, learn by reversing a noise-corruption process. A major challenge in discrete token spaces is that calculating the true reverse probability involves high-variance Monte Carlo estimates over the entire vocabulary. MDLM reveals that the token-level learning objective can be mathematically reformulated using Rao-Blackwellization—a statistical technique that reduces variance by conditioning on a subset of variables. Technically, this rewrites the complex diffusion objective into a mixture of standard masked cross-entropy losses. Instead of guessing blindly, the model computes a marginalized expectation over the forward noise process. This leads to a provably tighter Evidence Lower Bound (ELBO), meaning the model captures the true underlying token information density much more efficiently. As tokens undergo the reverse process, their semantic certainty increases continuously, effectively proving that masked diffusion is mathematically homologous to a continuous, iterative version of Masked Language Modeling (like BERT).
- **Cross-architecture Implications**
This is specific to masked diffusion objectives, though it conceptually bridges discrete diffusion math with the standard optimization objectives used in encoder-only Transformers
- **Limitations**
The paper strictly focuses on optimization objectives and probabilistic bounds. It does not explain the spatial interactions—meaning it doesn't analyze how Token A geometrically influences Token B within the hidden layers during a specific denoising step.

3.3 XDLM: Unified Masked + Uniform Diffusion

- **Architecture Studied**: Diffusion LLM
- **Token-level Object**: Transition probability matrices and noise kernels
- **Analytical Tool**: Information-Theoretic and Dynamical Systems (Markov transition dynamics)
- **Layer Wise vs Global**: Global (temporal trajectory)
- **Key Finding about Token Dynamics**
  In discrete language diffusion, tokens are usually corrupted in one of two distinct ways: Masked diffusion (replacing a token entirely with a [MASK] token) or Uniform diffusion (randomly flipping a token to another word in the vocabulary). XDLM discovers a theoretical stationary noise kernel that mathematically unifies both paradigms. By defining token corruption as a unified Markov transition matrix, the research shows that tokens transition through a predictable probability space where masking acts as an "absorbing state" and uniform flipping acts as an "ergodic mixing process." The key technical breakthrough is proving that the reverse-time dynamics (the actual generation process) can smoothly interpolate between these two extremes. Tokens can initially be retrieved from uniform noise (which establishes broad semantic and contextual domains) and later refined through mask-prediction mechanics (which locks in exact syntax and grammar).
- **Cross-architecture Implications**
  This uniquely solves a theoretical division within the Diffusion LLM family itself, creating a unified mathematical framework for discrete noise. It does not apply to autoregressive standard Transformers or SSMs.
- **Limitations**
  The mathematical proofs assume a fixed, distinct vocabulary space. It does not account for sub-word tokenization geometries (e.g., how the semantic distance between specific BPE subwords affects the optimal uniform transition matrix).

3.4 Inference-Time Scaling of Diffusion Language Models via Trajectory Refinement

- **Architecture Studied**: Diffusion LLM
- **Token-level Object**: Generation trajectory and posterior token distribution
- **Analytical Tool**: Dynamical Systems (Markov Chain Monte Carlo, Particle Gibbs sampling)
- **Layer Wise vs Global**: Global(trajectory-level)
- **Key Finding about Token Dynamics**
  A major mechanical flaw in standard diffusion language models is the accumulation of conditioning errors: if a token is incorrectly predicted early in the reverse-denoising path, that error severely corrupts the conditioning context for all surrounding tokens, leading to a degraded final sequence. PG-DLM models the entire sequence generation as a dynamic state-space trajectory. By applying Particle Gibbs (a sequential Monte Carlo method), the model runs multiple parallel token trajectories ("particles") and dynamically resamples them based on their likelihood scores at intermediate denoising steps. The technical achievement is providing strict theoretical convergence guarantees that this resampling actively pulls the token distribution back toward the true data manifold. Essentially, tokens are allowed to mathematically "change their minds" mid-generation if their current trajectory leads to low-probability semantic states, actively healing context errors before the final text is output
  Particle Gibbs sampling algorithm is an advanced Markov Chain Monte Carlo method that uses Sequential Monte Carlo or particle filtering to estimate complex probability distributions

- **Cross-architecture Implications**
While the continuous math is applied to Diffusion, the use of trajectory-level resampling conceptually mirrors search algorithms like best-first tree search or beam search used during Transformer inference.

- **Limitations**
Particle filtering is notoriously expensive computationally. The theoretical gains in token accuracy come at a massive latency cost during inference, making it difficult to scale to real-time text generation.

4 Mixture-of-Experts: Token Routing

4.1 DynaMoE: Dynamic Token-Level Expert Activation with Layer-Wise Adaptive Capacity for Mixture-of-Experts Neural Networks

- **Architecture Studied**: Mixture-of-Experts

- **Token-level Object**: Routing decision, expert activation count, and compute load per token.

- **Analytical Tool**: Routing/Gating

- **Layer Wise vs Global**: Layer-wise

- **Key Finding about Token Dynamics**
Standard MoE architectures enforce a rigid Top-K routing policy (e.g., routing every token to exactly 2 experts), which mechanically assumes every token requires the exact same amount of computational work to process. DynaMoE overturns this by analyzing the intrinsic "difficulty" of individual tokens. The paper implements a gating mechanism that dynamically allocates a variable number of experts based on a token's computed confidence threshold. Technically, simple syntactic tokens (like "the," "and," or punctuation) might be routed to a single expert or even bypass the MoE layer entirely (0 experts), while highly complex semantic tokens (like rare nouns, foreign words, or mathematical operators) trigger 4 or 5 experts simultaneously. This proves that token representations inherently contain a measurable "complexity gradient." By utilizing layer-wise adaptive capacity scheduling, the model prevents expert load imbalance while saving massive amounts of FLOPs on "easy" tokens, explicitly linking a token's information entropy to its computational routing requirement.

- **Cross-architecture Implications**
Highly MoE-specific. However, the underlying philosophy mirrors early-exit or variable-compute mechanisms sometimes grafted onto standard Transformers to save inference costs.

- **Limitations**
The paper leaves open a critical geometric question: if tokens use vastly different computational paths (some skipping layers, some using 5 experts), do their geometric positions in the final residual stream still align harmoniously, or does this dynamic routing lead to representation fragmentation?

4.2 Grassmannian Mixture-of-Experts: Concentration-Controlled Routing on Subspace Manifolds

- **Architecture Studied**: Mixture-of-Experts

- **Token-level Object**: Token masking probability and ELBO (Evidence Lower Bound) contributions

- **Analytical Tool**: (ELBO optimization, cross-entropy bounds)

- **Layer Wise vs Global**: Global (tracking tokens across temporal denoising steps)

- **Key Finding about Token Dynamics**
Traditional MoE routing uses a simple linear projection followed by a softmax to assign tokens to experts. This flat Euclidean approach suffers heavily from "expert collapse" (where all tokens flood the same few experts) and noisy, unstable routing decisions during training. This paper reformulates

routing as a purely geometric problem by mapping token representations and expert centroids onto a Grassmann manifold (a curved mathematical space parameterized by linear subspaces rather than points). Instead of standard softmax logits, tokens are routed based on the Bingham distribution, which naturally models antipodally symmetric probabilities on hyperspheres and Grassmannians. Technically, this geometric shift proves that expert collapse is largely an artifact of flat routing spaces. In the curved Grassmannian geometry, experts naturally repel each other into orthogonal subspaces. The paper provides formal mathematical bounds proving that under this geometry, routing entropy is perfectly preserved and expert collapse is mathematically impossible.

- **Cross-architecture Implications**
  While strictly analyzing MoE gating functions, it introduces powerful manifold-level analysis that could be used to study how dimensional reduction (like that seen in standard Transformers) impacts routing efficiency in hybrid models.

- **Limitations**
  Constraining the routing function to a Grassmannian manifold introduces complex Riemannian optimization during backpropagation. The paper primarily focuses on formal theoretical bounds rather than validating training stability at frontier scales (e.g., 70B+ parameters).

4.3 Mixture-of-Transformers (MoT)

- **Architecture Studied**: Mixture-of-Experts

- **Token-level Object**: Routing decision (expert assignment) and gradient geometry.

- **Analytical Tool**: Routing / Gating and Geometric / Manifold (focusing on subtask strong convexity).

- **Layer Wise vs Global**: Global (focuses on training dynamics and convergence).

- **Key Finding about Token Dynamics**
  This paper introduces a new MoE model that is designed to address the computational challenges of multi-modal model pretraining.
  Traditional dense models process all tokens through the same parameters, which often causes geometric "gradient conflicts" where updating weights to improve one token type degrades the optimization path for another. By routing tokens to specialized Transformer experts, the global, highly non-convex loss landscape is partitioned into distinct, strongly convex subtasks. This token specialization ensures that gradients from different semantic token clusters are orthogonal and do not interfere with one another, enabling an exponentially fast $O(\log(\varepsilon^{-1}))$ convergence rate for specialized semantic subsets.

- **Cross-architecture Implications**
  The principle that token-level specialization resolves gradient conflicts is conceptually relevant to Hybrid models that selectively route tokens to fundamentally different underlying mechanisms.

- **Limitations**
  The analysis heavily prioritizes convergence rates and optimization geometry; it does not detail how the intrinsic dimension or the Shannon entropy of the tokens changes sequentially as a result of passing through these experts.

4.4 DiffMoE: Dynamic Token Selection for Scalable Diffusion Transformers

- **Architecture Studied**: Mixture-of-Experts + Diffusion LLMs (Diffusion Transformers)

- **Token-level Object**: Routing decision across temporal denoising steps.

- **Analytical Tool**: Routing / Gating and Dynamical Systems

- **Layer Wise vs Global**: Global (tracks tokens temporally across the iterative generation process, rather than just spatially across layers)

- **Key Finding about Token Dynamics**
: In standard language diffusion, tokens are processed uniformly at every denoising timestep, which is wasteful because tokens resolve at different speeds. DiffMoE introduces a dynamic, token-level routing mechanism specifically adapted for this continuous dynamical system. Instead of independent layer-wise routing, it implements batch-level global token pools that continuously assess a token's confidence state. Tokens that achieve high certainty early in the reverse process are dynamically routed to "lighter" experts or dropped entirely, while highly uncertain tokens are routed to high-capacity experts. This proves that a token's computational requirement drastically decays as it moves along the diffusion trajectory.

- **Cross-architecture Implications**
This uniquely crosses the boundary between Diffusion and MoE, proving that MoE routing principles can be mapped onto the temporal axis of iterative denoising, not just the spatial depth of an autoregressive model.

- **Limitations**
The paper focuses primarily on computational load balancing and does not deeply analyze the information-theoretic bounds of how dropping tokens affects the strict mathematical guarantees of the diffusion reverse process.

5 Hybrid Architectures

5.1 Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning

- **Architecture Studied**: Hybrid (Mamba + Transformer + MoE)
- **Token-level Object**: Latent routing decisions
- **Analytical Tool**: Routing/Gating
- **Layer Wise vs Global**: Layer-wise (interleaved processing)
- **Key Finding about Token Dynamics**
Enforcing rigid boundaries between dense sequence processing and sequential processing limits token expressivity. This architecture implements "latent MoE," where the routing space is fundamentally expanded into a continuous latent space that seamlessly interacts with both Mamba state spaces and Transformer attention heads. The hybrid routing mechanism acts as an intelligent arbiter: tokens requiring long-context historical lookups are dynamically processed by sequential Mamba layers, while tokens requiring exact associative recall trigger dense self-attention layers. This dynamically optimizes sequence processing based on the specific semantic memory demands of each individual token.

- **Cross-architecture Implications**
Highly applicable to frontier models, demonstrating how the fundamental token processing mechanisms of Transformers, SSMs, and MoEs can be fused into a single unified routing framework.

- **Limitations**
As an applied frontier model paper, it relies heavily on empirical scaling results and lacks the rigorous mathematical proofs (like circuit complexity bounds or strict ODE formulations) explaining the fundamental theory of why tokens self-organize this way.

5.2 Jamba-1.5: Hybrid Transformer-Mamba Models at Scale

- **Architecture Studied**: Hybrid (interleaved Mamba + Transformer + MoE layers)
- **Token-level Object**: Memory retention and computational pathway assignment
- **Analytical Tool**: Routing / Gating and Information-Theoretic

- **Layer Wise vs Global**: Layer-wise (tracking tokens as they pass through heterogeneous blocks)
- **Key Finding about Token Dynamics**
  By interleaving Mamba layers with Transformer layers, the token sequence is subjected to an alternating computation rhythm. During Mamba layers, tokens aggressively filter information by compressing historical context into a fixed-size hidden state. To counter the exact-retrieval limitations of this compressed state, tokens are periodically passed through explicit self-attention layers. The Mamba layers act as a highly efficient background context synthesizer, while the intermittent Transformer layers act as sharp, precise memory retrieval mechanisms. This alternating flow proves that tokens do not need to carry exact pairwise attention scores at every layer; computing dense attention sparsely is sufficient to maintain global sequence coherence and bypass the information bottleneck of pure state spaces
- **Cross-architecture Implications**
  Validates the hypothesis that sequential state-mediated processing and dense all-to-all processing are complementary, rather than mutually exclusive, ways to evolve token representations
- **Limitations**
  The paper does not provide a granular geometric analysis of how a token's intrinsic dimension or manifold shape expands and contracts as it abruptly shifts from a Mamba block into a Transformer block